

Lab 2 — INC-0087 — The Storage Nobody Can Access

RBAC · Private Endpoint · Private DNS · Storage Firewall · Service Endpoints

ENVIRONMENT

Subscription	Subscription 2026 Alan
Resource Group	lab2-rg
Location	canadacentral
Storage Account	lab2sace7exhub5u5j6
VNet	lab2-vnet (10.1.0.0/16)
Private Endpoint	lab2-pe → blob target
Private DNS Zone	privatelink.blob.core.windows.net
Tools	PowerShell 7.6.0 · Azure CLI · Bicep

PHASE 1 — DEPLOY

1 Get Azure AD Object ID

PASS

```
# Required for RBAC role assignment in Bicep
az ad signed-in-user show --query id -o tsv
```

```
e895ea1e-f578-4a12-ba8a-65d0f4cd1216
```

2 Create Resource Group

PASS

```
az group create --name lab2-rg --location canadacentral
```

3

Deploy Bicep Template

PASS

```
# Deployed: NSG, VNet, Storage Account, Private Endpoint, DNS Zone, Role Assignment
az deployment group create `
  --resource-group lab2-rg `
  --template-file main.bicep `
  --parameters currentUserObjectId="e895ea1e-f578-4a12-ba8a-65d0f4cd1216"
  --name lab2-deployment
```

```
"provisioningState": "Succeeded"
"storageAccountName": "lab2sace7exhub5u5j6"
```

4

Get Storage Account Name

PASS

```
az storage account list --resource-group lab2-rg --query "[].name" -o tsv
```

```
lab2sace7exhub5u5j6
```

PHASE 2 — DIAGNOSE · PROBLEM 1: RBAC

5

Check Role Assignments on Storage Account

PROBLEM

```
# Azure CLI - scoped to storage account
az role assignment list `
  --scope $(az storage account show -g lab2-rg -n lab2sace7exhub5u5j6 --
query id -o tsv) `
  --query "[].{Role:roleDefinitionName, Principal:principalId}" `
  -o table
```

Role	Principal
Storage Account Contributor	e895ea1e-f578-4a12-ba8a-65d0f4cd1216

● PROBLEM CONFIRMED

Role is "Storage Account Contributor" — manages account settings only. Cannot read or write blob data. Needs "Storage Blob Data Contributor" for blob operations.

PHASE 3 — FIX · PROBLEM 1: RBAC

6 Remove Wrong Role

PASS

```
az role assignment delete `
  --assignee "e895eale-f578-4a12-ba8a-65d0f4cd1216" `
  --role "Storage Account Contributor" `
  --scope $(az storage account show -g lab2-rg -n lab2sace7exhub5u5j6 --
query id -o tsv)
```

7 Assign Correct Role

PASS

```
az role assignment create `
  --assignee "e895eale-f578-4a12-ba8a-65d0f4cd1216" `
  --role "Storage Blob Data Contributor" `
  --scope $(az storage account show -g lab2-rg -n lab2sace7exhub5u5j6 --
query id -o tsv)
```

8 Verify Role Fix

PASS

```
az role assignment list `
  --scope $(az storage account show -g lab2-rg -n lab2sace7exhub5u5j6 --
query id -o tsv) `
  --query "[].{Role:roleDefinitionName, Principal:principalId}" `
  -o table
```

Role	Principal
Storage Blob Data Contributor	e895eale-f578-4a12-ba8a-65d0f4cd1216

✓ **CONFIRMED OK**

Role correctly updated to Storage Blob Data Contributor. Blob read/write access granted.

PHASE 4 — DIAGNOSE · PROBLEM 2: DNS

9 Check VNet Link on Private DNS Zone

PROBLEM

```
az network private-dns link vnet list `
```

```
--resource-group lab2-rg \  
--zone-name "privatelink.blob.core.windows.net" \  
-o table
```

(empty — no results)

● PROBLEM CONFIRMED

No VNet link exists on the Private DNS Zone. VMs inside lab2-vnet cannot use the private DNS directory — they fall back to public DNS and get the storage public IP.

10 Check DNS Zone Group on Private Endpoint

PROBLEM

```
az network private-endpoint dns-zone-group list \  
--resource-group lab2-rg \  
--endpoint-name lab2-pe \  
-o table
```

(empty — no results)

● PROBLEM CONFIRMED

No DNS Zone Group on the private endpoint. The private IP (10.1.2.4) is never auto-registered in the DNS zone. Storage hostname resolves to public IP instead.

PHASE 5 — FIX · PROBLEM 2: DNS

11 Create VNet Link on Private DNS Zone

PASS

```
az network private-dns link vnet create \  
--resource-group lab2-rg \  
--zone-name "privatelink.blob.core.windows.net" \  
--name lab2-dns-link \  
--virtual-network lab2-vnet \  
--registration-enabled false
```

```
"provisioningState": "Succeeded"  
"virtualNetworkLinkState": "Completed"
```

12

Create DNS Zone Group on Private Endpoint

PASS

```
az network private-endpoint dns-zone-group create `
  --resource-group lab2-rg `
  --endpoint-name lab2-pe `
  --name lab2-dns-zone-group `
  --private-dns-zone "privatelink.blob.core.windows.net" `
  --zone-name blob
```

```
"fqdn": "lab2sace7exhub5u5j6.privatelink.blob.core.windows.net"
"ipAddresses": ["10.1.2.4"]
"provisioningState": "Succeeded"
```

 CONFIRMED OK

DNS Zone Group created. Private IP 10.1.2.4 auto-registered. Storage hostname now resolves to private IP inside the VNet.

PHASE 6 — DIAGNOSE · PROBLEM 3: FIREWALL

13

Check Storage Firewall Rules

PROBLEM

```
az storage account show `
  --resource-group lab2-rg `
  --name lab2sace7exhub5u5j6 `
  --query "networkRuleSet" -o json
```

```
{
  "bypass": "AzureServices",
  "defaultAction": "Deny",
  "ipRules": [],
  "virtualNetworkRules": []
}
```

 PROBLEM CONFIRMED

defaultAction is Deny. virtualNetworkRules is empty. All traffic blocked including VNet subnet traffic.

PHASE 7 — FIX · PROBLEM 3: FIREWALL

14 Enable Service Endpoint on Subnet

PASS

Required before subnet can be added to storage firewall rules.

```
az network vnet subnet update `
  --resource-group lab2-rg `
  --vnet-name lab2-vnet `
  --name lab2-pe-subnet `
  --service-endpoints Microsoft.Storage
```

15 Add Subnet to Storage Firewall Rules

PASS

```
az storage account network-rule add `
  --resource-group lab2-rg `
  --account-name lab2sace7exhub5u5j6 `
  --vnet-name lab2-vnet `
  --subnet lab2-pe-subnet
```

```
"virtualNetworkRules": [
  {
    "action": "Allow",
    "state": "Succeeded",
    "virtualNetworkResourceId": "...lab2-vnet/subnets/lab2-pe-subnet"
  }
]
```

✓ CONFIRMED OK

lab2-pe-subnet added to storage firewall with action Allow. VNet traffic now permitted.

PHASE 8 — END-TO-END VERIFICATION

16 Add Local IP for Test Upload

PASS

```
# Get public IP of local machine
(Invoke-WebRequest -Uri "https://api.ipify.org").Content

# Add to storage firewall
az storage account network-rule add `
  --resource-group lab2-rg `
```

```
--account-name lab2sace7exhub5u5j6 `
--ip-address 216.131.118.164
```

17 Create Blob Container

PASS

```
az storage container create `
  --account-name lab2sace7exhub5u5j6 `
  --name test `
  --auth-mode login
```

```
"created": true
```

18 Upload Test Blob — Final Verification

PASS

```
echo "lab2 test" > testfile.txt

az storage blob upload `
  --account-name lab2sace7exhub5u5j6 `
  --container-name test `
  --name testfile.txt `
  --file testfile.txt `
  --auth-mode login
```

```
Finished[#####] 100.0000%
"request_server_encrypted": true
"lastModified": "2026-05-14T01:02:57+00:00"
```

✓ CONFIRMED OK

Blob uploaded successfully at 100%. Server-side encryption confirmed. All three problems resolved. INC-0087 closed.

PHASE 9 — CLEANUP

```
# Remove all resources to avoid charges
az group delete --name lab2-rg --yes --no-wait
```

COMMAND REFERENCE — QUICK LOOKUP

Purpose	Command
List role assignments on resource	<code>az role assignment list --scope <resource_id></code>
Remove a role	<code>az role assignment delete --assignee <id> --role "<name>"</code>
Assign a role	<code>az role assignment create --assignee <id> --role "<name>"</code>
Check DNS zone VNet links	<code>az network private-dns link vnet list --zone-name <zone></code>
Create VNet link on DNS zone	<code>az network private-dns link vnet create --virtual-network <vnet></code>
Check DNS zone groups on PE	<code>az network private-endpoint dns-zone-group list --endpoint-name <pe></code>
Create DNS zone group on PE	<code>az network private-endpoint dns-zone-group create</code>
Check storage firewall	<code>az storage account show --query "networkRuleSet"</code>
Enable service endpoint on subnet	<code>az network vnet subnet update --service-endpoints Microsoft.Storage</code>
Add subnet to storage firewall	<code>az storage account network-rule add --subnet <subnet></code>
Add IP to storage firewall	<code>az storage account network-rule add --ip-address <ip></code>
Upload blob via CLI	<code>az storage blob upload --auth-mode login</code>
Get your public IP	<code>(Invoke-WebRequest -Uri "https://api.ipify.org").Content</code>
List all resources in RG	<code>az resource list --resource-group <rg> --output table</code>